WORKSHEET

Chapter 9: Exponential distributions

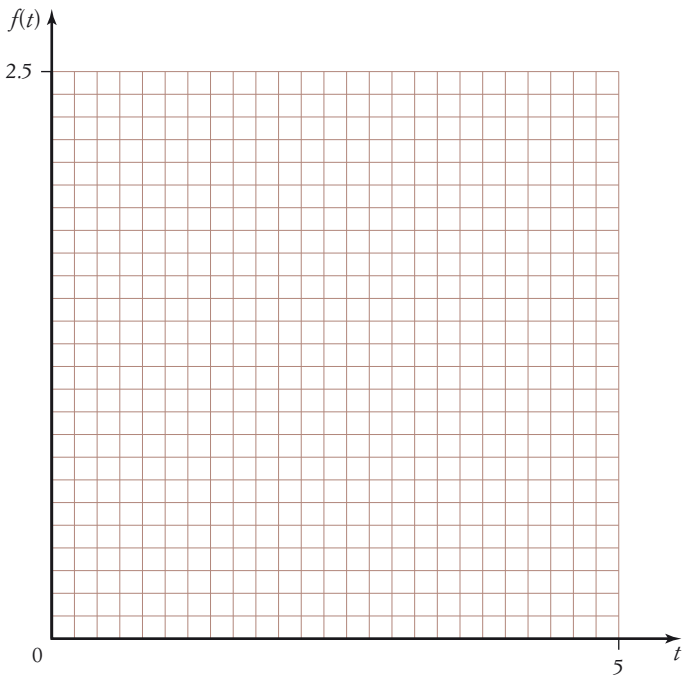
1 Draw graphs of the exponential distribution $f(t) = \lambda e^{-\lambda t}$ for the following values of λ on the same set of axes shown below.

a $\lambda = 2$

b $\lambda = 1$

c $\lambda = \frac{1}{5}$

d $\lambda = \frac{1}{20}$



2 Compare the shapes of the distributions graphed in Question **1**.

3 The expected life (in hours) of an electronic component can be considered to be a random variable with an exponential distribution for which $\lambda = \frac{1}{400}$.

a Find the probability that a particular component lasts at least 500 h.

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b Find the expected time that a particular component lasts.

4 The average life of an industrial safety switch is known to be 1.8 years. Use an exponential distribution to model the life of safety switches to find answers, correct to 5 decimal places, for

a λ

b the proportion of switches that fail before 1.8 years

c the proportion of switches that last for at least 5 years.

5 A random variable X has an exponential distribution with $\alpha = 0.4$. Find each of the following.

a $P(0 \leq x \leq 3)$

b $P(3 \leq x \leq 10)$

c $P(9.5 \leq x \leq 10.5)$

d $P(x > 10)$

6 A random variable X has an exponential distribution with $\alpha = 2$. Find each of the following.

a $P(0 \leq x \leq 3)$

b $P(0.3 \leq x \leq 0.7)$

c $P(x > 0.3)$

d $P(x > 0.8)$

7 An exponential distribution has parameter $\lambda = 0.0025$. What is the expected value of the distribution?

8 An exponential distribution has parameter $\lambda = 0.0005$. What is the expected value of the distribution?

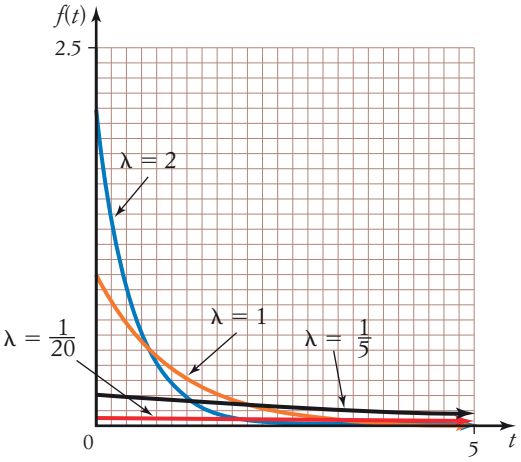
9 What is the parameter λ for an exponential distribution with an average value of 12?

10 What is the parameter λ for an exponential distribution with an average value of 25?

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Answers

1



2 Higher values of λ start higher and are skewed more to the left, so these distributions have a higher probability of a lower value of t.

3 **a** About 0.2865

b 400 h

4 **a** 0.55556 (or $\frac{5}{9}$)

b 0.63212

c 0.06208

5 **a** 0.69881

b 0.28288

c 0.00738

d 0.01832

6 **a** 0.99752

b 0.30221

c 0.54881

d 0.2019

7 400

8 2000

9 $\frac{1}{12}$

10 0.04

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